

PhD Project Outline

Decision-Oriented Information Acquisition for Private-Market Investment Decisions: When Does Acquiring the Next Signal Pay for Itself?

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Target institution

ETH Zurich, Department of Computer Science (D-INFK) — Learning & Adaptive Systems Group (Prof. Andreas Krause), Institute for Machine Learning. A domain co-advisor at HSG (private markets / entrepreneurial finance) is envisaged.

1. Research context

An investor deciding under uncertainty does not buy a prediction. They buy **information** that changes a decision. Yet the dominant AI framing in private markets is predictive: forecast which company succeeds. That framing is crowded and, taken alone, answers the wrong question — it improves an estimate without telling anyone what to go and find out next.

The prior question is decision-oriented: **given the current information state and a cost budget, which information block — if any — should be acquired before continuing or stopping diligence?**

This is not a new question in the abstract. It is precisely the object of your group's work on optimising value of information, of *Near-Optimal Bayesian Active Learning for Decision Making* (Javdani et al., AISTATS 2014), and of adaptive submodularity as a structural condition under which adaptive greedy acquisition is provably near-optimal (Golovin & Krause, JAIR 2011). Yuxin Chen's ETH thesis, *Near-optimal Adaptive Information Acquisition* (2017), is the closest existing statement of the machinery.

What is open is the domain. That apparatus has been validated on active learning, troubleshooting, object detection, sensor placement and viral marketing — settings with cheap, repeatable, largely independent tests. Private-market investment decisions violate almost every one of those conveniences: tests are expensive and slow, correlated, partly unobservable, and the outcome arrives years later under confounding. My claim is not that the theory needs replacing. It is that **this is the setting in which its assumptions bind hardest, and nobody has checked what happens there.**

2. Research questions

Main question: In investment decisions under uncertainty, which information acquisition — at what cost and delay — most improves decision quality, and when should acquisition stop?

- How should the *information state* $\mathcal{I}_t$ be defined so that it contains only what was defensibly observable at time t , with no look-ahead?
- Do the structural conditions that license near-optimal adaptive greedy acquisition (adaptive submodularity, or a usable surrogate) **hold, approximately hold, or fail** when tests are costly, correlated and noisy in the way diligence steps are?
- Under what cost structure does a decision-aware acquisition policy beat cheaper heuristics on **net decision value** — and under what cost structure does it provably not?

That third question is where my current evidence points, and I state it as a question rather than a claim because my own result so far is negative (see §3).

3. Foundation and honest evidence boundary

Completed work. *Can Non-Financial Signals Price Private Companies?* (Guidi, Rashid & Zhong; accepted, WI2026 Student Track, Business Informatics). On 3,403 PitchBook deals under a strict out-of-time holdout spanning the 2022 regime shift, non-financial signals priced private companies competitively

with the strongest financial baseline, and investor-syndicate capacity was the most influential signal among those tested. The value of that result here is diagnostic: it maps the ceiling of the predictive framing and motivates the shift to acquisition.

Current state of the flagship — stated plainly. I have built a point-in-time pipeline on public SEC data: US technology-related primary issuers with non-amendment Form D anchors; decision time 12 months after filing availability; company-disjoint temporal splits (development 2021, validation 2022, locked test 2023 untouched).

- A real SEC-history information block is **predictive**: the frozen baseline raises ROC-AUC from 0.5930 to 0.6551 on the 2022 validation cohort, and transports across cohorts.
- The **cost-aware value-of-information policy currently fails** to beat the preregistered non-VoI baselines on net decision value: it passes 0 of 15 declared utility-by-cost scenarios, with a best point estimate of +0.00340 whose bootstrap interval crosses zero. Costs were not adjusted after seeing that result. This is an exploratory negative result, not a published finding.
- The outcome label is an explicitly **weak proxy** (a later non-amendment Form D notice within 18 months) — not a priced round, Series A, or success label. A 20-case model-reviewed pilot reached 75% raw exact agreement with 45% initially unresolved; I treat it as adequate for protocol development and sensitivity analysis only, **not** as human-adjudicated gold data.

I lead with the negative result deliberately. Either the utility model is mis-specified, or the cost structure of public information makes acquisition genuinely not worth it in this regime, or the structural conditions for decision-oriented greedy acquisition do not hold here. **Which of those three it is, is a research question worth a thesis, and it is a methodological question, not a finance one.**

4. Methodology

Feasibility does not depend on proprietary data.

- **Point-in-time reconstruction** from dated public evidence defines I_t and excludes look-ahead. Proprietary sources (PitchBook, Orbis) are not a dependency; they may later enter *as acquirable blocks whose marginal value is estimated* — so "is this paid signal worth its cost?" stays a research question rather than a precondition.
- **Diagnosing the negative result**: separate mis-specified utility from genuinely unfavourable cost structure from failure of the structural conditions. This is the Year-1 core and the part that most needs your group's theory.
- **A synthetic multi-world harness** where the data-generating process is sealed from the tested policy, used for recovery and counterfactual checks. It validates plumbing and method recovery; it does *not* establish market realism, and I will not present it as evidence of real-world performance.
- **Preregistered baselines** — acquire-all, acquire-none, cheapest-first, uncertainty-based, decision-aware VoI — scored on decision value net of information cost and delay, never on prediction accuracy alone. The 2023 cohort stays locked.
- **Year 1**: formalise the acquisition objective and stopping rule; diagnose the negative result; test whether a usable structural condition or surrogate exists in this setting.
- **Year 2**: completion and submission. (If it grows: how to judge whether a decision was good given what was knowable, separately from how it turned out.)
- **Year 3–4**: robustness, correlated and partially observable tests, writing.

5. Expected contribution

(i) A **stress test of decision-oriented information acquisition** in a domain where its standard assumptions are violated — costly, delayed, correlated, partly unobservable tests with confounded long-horizon outcomes; (ii) a **reproducible point-in-time methodology** that makes the question testable on public data, with the evidence boundary stated rather than blurred; (iii) a characterisation of **when**

cost-aware acquisition does and does not pay, which is domain-general — the same formulation applies to credit, M&A, and clinical or defence triage. Private markets are the entry domain, not the limit.

6. Fit with the group

Your group's stated focus is systems that "actively acquire information, reason and reliably make decisions in complex and uncertain domains." My project is that sentence applied to a domain the group has not worked in, and it arrives with the infrastructure already built and a falsified first attempt rather than a proposal.

I am aware the group's current output centres on foundation models, flow-based generative optimisation, test-time training and reinforcement learning rather than the 2010–2017 value-of-information line. I read that as an opportunity rather than a mismatch: the recent work on **active data selection** — deciding which data is worth acquiring at test time — is the same question under a modern name, and my setting supplies an application where acquisition cost is real money and real delay rather than compute.

What I bring is not theoretical depth, which is what I am asking to acquire. It is a built point-in-time data layer, a strict evaluation discipline, and a demonstrated willingness to report my own negative result before someone else finds it. My degrees are in management and my professional roles have been in product management (currently Amazon Business, previously consulting at Capgemini), so the empirical and engineering work is self-taught and self-directed.

7. Selected references

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- Ma, C., et al. (2019). *EDDI: Efficient Dynamic Discovery of High-Value Information*. ICML.
- Fu, H., & Taylor, L. (2024). *Due Diligence and the Allocation of Venture Capital*. SSRN 5014747.
- Gompers, P., Gornall, W., Kaplan, S., & Strebulaev, I. (2020). *How Do Venture Capitalists Make Decisions?* Journal of Financial Economics.
- Guidi, A., Rashid, S., & Zhong, H. (2026). *Can Non-Financial Signals Price Private Companies?* WI2026 Student Track (accepted).

One-page WI2026 summary attached. Code, point-in-time pipeline and manuscript available on request.